
SENPAI: Self-Experimentation for Physical AI

An Observability-Based Research Harness

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Abstract

SENPAI (Self-Experimentation for Physical AI) is an observability-first research harness in which multi-agent state is grounded in pull requests and structured experiment logs rather than agent memory and scratchpads. Although task-agnostic, we evaluate SENPAI on CFD-surrogate recipe search, where performance depends on architecture, optimization, losses, normalization, and physics-aware considerations. SENPAI uses a thin Advisor/Student loop to turn hypotheses into PRs, training runs, and PR comments, producing an experiment ledger queryable by both agents and researchers. This makes experiments auditable, recoverable, and human-steerable. We evaluate Transolver-based SENPAI across DrivAerML, AirfRANS, and Tandem-FoilSet. Starting from the original Transolver model, SENPAI improves DrivAerML surface-pressure and wall-shear-stress relative- L_2 error, reduces AirfRANS surface-MSE below all compared references, and reaches lower Tandem-FoilSet cruise-random-uniform normalized full-field MSE than the reported benchmark. Overall, SENPAI supports sparsely steered, multi-day semi-autonomous research that can deliver strong task-specific models. We will release the harness and experiment ledger, including hosted experiment logs, so the research programme can be audited end to end.

1. Introduction

CFD surrogates offer a way to accelerate engineering design: learned models can screen early-stage geometries, explore operating regimes, and focus expensive CFD or physical testing on the most promising candidates. Recent surrogate models now handle larger meshes, irregular geome-

tries, stricter accuracy targets, and smaller training sets than were feasible only a few years ago (Zhou et al., 2026), making hybrid ML-CFD workflows increasingly attractive.

In practice, however, training a useful CFD surrogate is still a niche capability. Performance depends on architecture, optimization, normalization, boundary conditions, symmetries, and rollout stability (Wang et al., 2024). Strong recipes require both ML expertise and domain knowledge in aerodynamics or fluid dynamics (Hodges, 2024), which many engineering teams do not have. SENPAI targets this gap by exploring, documenting, and refining problem-specific surrogate recipes while keeping researchers in the loop.

Recent autonomous-research systems show that LLM agents can search literature, propose hypotheses, edit code, run experiments, and produce reports (Lu et al., 2024; Schmidgall et al., 2025; Yamada et al., 2025; Miyai et al., 2026). SENPAI builds on this direction but focuses on a different systems question: how should long-running, multi-experiment ML research be represented so that both agents and researchers can audit, recover, compare, and steer it?

SENPAI’s central design choice is to ground the research loop in external systems of record that ML teams already inspect: pull requests and git history for hypotheses, code changes, review, and provenance (GitHub Docs, 2026; Chacon & Straub, 2014); and a hosted experiment tracker for configurations, metrics, checkpoints, system telemetry, and agent traces (Biewald, 2020; Weights & Biases, 2026). Our implementation uses GitHub and Weights & Biases,¹ but the harness design is not specific to either service. Each hypothesis, code diff, training result, review decision, failed run, and baseline update becomes both machine-queryable and research-reviewable through these existing tools.

The harness loop is deliberately thin. An Advisor reads the current experiment ledger, proposes a literature-grounded hypothesis as a draft PR, and assigns it to a GPU-backed Student. The Student checks out the branch, modifies the training code, runs the experiment, logs metrics to the experiment tracker, and reports results in the PR. The Advisor

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¹Weights & Biases was the hosted experiment tracker used in the reported experiments.

then merges, requests revision, or closes the line of inquiry. Humans can steer the Advisor or Student through GitHub issues and PR comments, while markdown scratchpads remain secondary sources of state rather than the primary system of record.

Across the research programme, SENPAI produced over 3,000 PRs, 11,000 experiment-tracker runs, and operated with up to 59 concurrent Student agents, turning a multi-week semi-autonomous research programme into a durable experiment ledger.

Contributions.

- **Observable research state.** We introduce SENPAI, a semi-autonomous ML research harness that grounds experiment state in PRs, commits, and experiment-tracker runs, producing a ledger that is queryable by agents and inspectable by researchers.
- **CFD recipe search.** Starting from Transolver-style baselines, SENPAI discovers improved recipes across AirFRANS, DrivAerML, and TandemFoilSet, with strongest gains on surface quantities.
- **Failure modes at scale.** We analyze a 24-hour fleet trace with 53,022 Claude requests and 5.24B tokens, showing that the dominant failures were monitor-driven context bloat, brittle tool interfaces, checkpoint/result-capture gaps, and state reconciliation issues rather than failures of scientific reasoning.

2. Related Work

Recent autonomous-research systems show that LLM agents can search literature, propose hypotheses, edit code, run experiments, and produce research reports. The AI Scientist generates ideas, implements experiments, writes papers, and runs automated review (Lu et al., 2024; Sakana AI, 2024), while AI Scientist-v2 extends this direction with agentic tree search and workshop-level paper generation (Yamada et al., 2025; Sakana AI, 2025). Agent Laboratory structures literature review, experimentation, and report writing around a researcher-provided idea (Schmidgall et al., 2025), and Jr. AI Scientist iteratively improves a baseline paper using modern coding agents (Miyai et al., 2026). Related systems emphasize collaboration, shared artifacts, or domain tools: ResearchAgent generates and revises ideas over scientific-literature graphs (Baek et al., 2024), AgentRxiv lets agent laboratories upload and retrieve generated reports from a shared preprint server (Schmidgall & Moor, 2025), ChemCrow and Coscientist connect LLM planners to chemistry tools and laboratory workflows (Bran et al., 2024; Boiko et al., 2023), and FunSearch couples an LLM generator to a verifiable evaluator and scored program database (Romera-Paredes et al., 2024). Similarly, in

general ML engineering, Hugging Face’s ml-intern is a recent, non-science-focused example that reviews papers, writes training code, and runs experiments (Hugging Face, 2026). These systems establish the value of tool use, execution, and persistent artifacts in autonomous research. SENPAI builds on that direction, but studies a different substrate question: how should the state of a long-running, multi-experiment ML research programme be represented so that both agents and researchers can inspect, compare, recover, and steer it?

The closest systems to SENPAI also recognize that long-horizon agents need state outside the model context, but they choose different substrates. Kosmos maintains an internal structured memory of literature-search and data-analysis summaries to choose later tasks, and cites final report claims with papers or generated Jupyter notebooks (Mitchener et al., 2025). AiScientist uses a permission-scoped File-as-Bus workspace so agents can re-ground on file-based analyses, plans, code, logs, and experimental evidence (Chen et al., 2026). Grounded autonomous research externalizes state as rerunnable simulation artifacts and comparison results within a single-paper reproduction loop (Huang, 2026). SENPAI takes a complementary systems position: the authoritative state should be the MLOps tooling machine learning researchers already use. In SENPAI, pull requests carry hypotheses, code review, discussion, and review decisions (GitHub Docs, 2026); git history provides ordered code provenance (Chacon & Straub, 2014); and hosted experiment trackers provide structured records of metrics, hyperparameters, artifacts, checkpoints, system telemetry, and traces (Biewald, 2020; Weights & Biases, 2026). This distinction matters most at scale: once an autonomous research system produces hundreds or thousands of experiments, the central challenge is no longer only whether an agent can run an effective experiment, but whether researchers can understand the programme it is creating.

3. System Design

Core principle. SENPAI’s central design principle is that authoritative experiment state lives in pull requests, git history, and experiment-tracker runs, not in agent memory or local scratchpads. The harness is therefore organized around making each hypothesis, code change, training result, and review decision durable, queryable, and inspectable through existing research tools.

PR-mediated lifecycle. SENPAI is a thin two-role harness over a task repository. An Advisor reads the current experiment ledger, optionally calls a literature-search sub-agent, opens draft PRs containing hypotheses and experiment instructions, and assigns them to GPU-backed Stu-

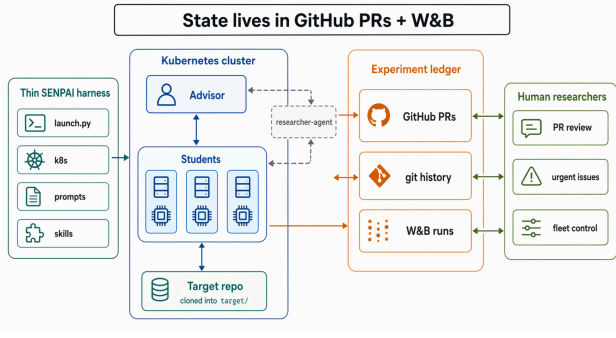


Figure 1. SENPAI deployment architecture. A thin harness deploys one Advisor and N Student pods; both roles can call a shared literature-search sub-agent. All programme state lives outside the cluster in the experiment ledger: PRs, git history, and experiment-tracker runs.

dents. Each Student claims its labeled PR, edits the branch, launches training with experiment-tracker logging, and reports results back as PR comments. Researchers can steer Advisors or Students via labeled GitHub issues or by leaving comments on a PR. A hypothesis’s trajectory therefore remains visible as a standard PR history linked to experiment-tracker runs, rather than as an agent-internal trace.

Design choices. SENPAI follows three design choices: externalizing experiment state into existing systems of record, keeping the autonomous loop thin, and exposing the same ledger to both agents and researchers. The role prompts, problem-program contract, helper skills, and outer-loop pseudocode are summarized in Appendices A to D.

3.1. Externalized Experiment State

The experiment ledger has two linked records. The PR records the hypothesis, implementation instructions, code diff, Student report, Advisor review, and final decision. The experiment tracker records the run details: configuration, metrics, checkpoints, system data, and agent traces. In our experiments, this tracker was Weights & Biases. Together these records are inspectable through familiar code-review and experiment-tracking interfaces and queryable through mature CLIs, allowing both researchers and agents to audit individual experiments or summarize programme-level progress.

The Advisor also maintains lightweight markdown summaries for focus, such as current baseline metrics, dataset notes, and open research directions. These files are treated as caches rather than authoritative state: when summaries disagree with PRs, git history, or experiment-tracker runs, the ledger wins.

3.2. Thin Advisor/Student Loop

Each experiment follows a PR-mediated lifecycle. The Advisor drafts a PR with a hypothesis, implementation guidance, target metrics, baseline values, and a Student label. The Student implements the change on the PR branch, runs training with experiment-tracker logging, and posts commands, training run links, metrics, and experiment interpretation as a PR comment. Once the PR is marked ready for review, the Advisor compares the result against the stated baseline and either merges the improvement, requests revision, or closes the branch. Merged PRs advance the programme baseline used for subsequent hypotheses.

SENPAI deploys as a small set of Kubernetes workloads: one Advisor CPU pod, N Student GPU pods, and a shared persistent data volume for datasets and checkpoints. The harness image is stable across problems; a task repository can be swapped in per SENPAI research programme, so every agent commit, branch, and PR lives in a self-contained target repository that a researcher can inspect independent of the harness.

The outer harness loop is programmatic rather than agentic. A shell wrapper polls GitHub and Student state for review-ready PRs, human issues, new comments, and idle Students; if no action is available, it sleeps without invoking the LLM. When action is required, the wrapper starts or resumes the Advisor with a triage message. This keeps routine polling outside the model context and focuses the agent on research decisions rather than heartbeat bookkeeping.

3.3. Human Steering Through Shared Observability

Researchers primarily steer SENPAI’s Advisor or Students through GitHub issues, while PR comments can also provide experiment-level intervention. This channel is intentionally low bandwidth: SENPAI is designed for sparse steering rather than continuous supervision, and each intervention remains attached to the public experiment record it affected.

4. Experiments and Results

4.1. Experiment Setup

Harness. Claude Code (v2.1.85 to v2.1.117) was run in headless mode as the agent harness for both the Advisor and Students. Claude Opus 4.6 and Claude Opus 4.7 were used with a compaction trigger at 200k tokens. Student pods were scheduled on NVIDIA RTX 6000 Blackwell nodes with 96 GB VRAM per GPU, from 1 to 8 GPUs per Student node. At maximum, $N = 59$ concurrent Students were tested, with peak concurrent training runs of 347. Over the five-week experimentation and harness-development cycle, 30 billion tokens were processed by Claude Code while run-

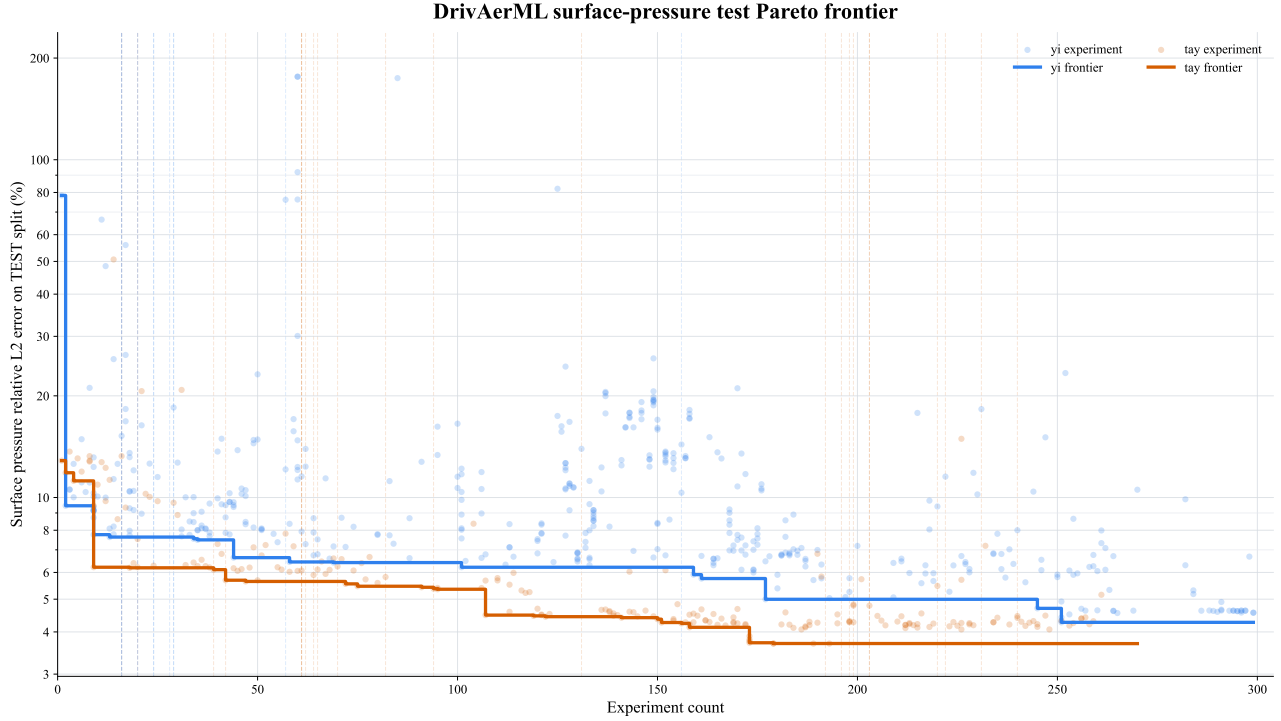


Figure 2. Evolution of DrivAerML surface-pressure test error across two SENPAI campaigns. Faded markers show individual completed experiments from the `yi` and `tay` branches, while solid curves show the cumulative Pareto frontier: the best surface-pressure relative L_2 error achieved up to each experiment count. The vertical lines mark GitHub-recorded human researcher/operator interventions. `tay` was instructed to use DDP8 and was given a 6 hour per-experiment training budget, hence the better performance.

ning physical-AI experiments.

Scale. Over the course of the research programme, SENPAI created over 3,000 PRs and recorded over 11,000 experiment-tracker runs while deploying, at peak, 59 distinct Student agents. Approximately 30 billion Claude Code tokens were processed during experimentation.

4.2. Datasets and Metrics

To demonstrate SENPAI’s CFD performance across multiple benchmarks, we train and evaluate on three CFD surrogate datasets spanning 2D tandem-airfoil flow, 2D airfoil RANS, and 3D automotive aerodynamics: TandemFoilSet (Lim et al., 2026), AirfRANS (Bonnet et al., 2022), and DrivAerML (Ashton et al., 2024).

TandemFoilSet. We use the paper’s original Experiment 4 partition, Cruise Random and Race Car tasks, as one of two TandemFoilSet datasets and measure denormalized surface-pressure MAE. In addition, we introduce a second dataset split, TandemFoilSet-Balanced (Anonymous, 2026b), with 1,499 training cases across four balanced test splits: one single-foil in-distribution split, two unseen-front-camber geometry holdouts (race-car and cruise), and one Reynolds-number holdout. An equal-weight average denor-

malized surface-pressure MAE of these four splits is used as the test metric.

AirfRANS. We use the official full task with the last 10% of the official training list held out for validation, giving a 720/80/200 train/validation/test split while keeping the official test set unchanged. We test on normalized targets on the official test split.

DrivAerML. We use the public surface split on the available processed cases, 400/34/50 train/validation/test, and measure surface-pressure relative- L_2 in percent, computed per case on unnormalized predictions and targets and then averaged over test cases.

4.3. DrivAerML

Result. On the DrivAerML benchmark, SENPAI, starting from a Transolver-based implementation, improves on the original Transolver’s test-set metrics of surface-pressure relative- L_2 error (3.98% versus 4.81%) and vector wall-shear-stress relative- L_2 error (7.31% versus 8.95%). Figure 2 shows the autonomous experiment trajectory for surface-pressure test error across the main DrivAerML branches.

Table 1. DrivAerML test relative- L_2 error (%). Lower is better.

Method	p_s	τ	p_v
SENPAI, ensemble	3.69	6.86	11.36
Transolver-3 (Zhou et al., 2026)	3.71	5.85	5.72
AB-UPT (Alkin et al., 2025)	3.82	7.29	6.08
SENPAI, single model	3.98	7.31	11.33
Transolver++	4.12	6.42	6.70
Transolver (Wu et al., 2024)	4.81	8.95	7.74

Recipe. The final training recipe SENPAI found used the original Transolver backbone but scaled it to 4 layers, 512 hidden units, and 128 slices, replaced fixed sinusoidal coordinates with a separable learnable Fourier positional embedding initialized across multiple spatial scales (Li et al., 2021; Schenck et al., 2025), and trained with Lion (Chen et al., 2023) plus EMA. It also added physics-preserving y -plane reflection augmentation, flipping the lateral coordinate, surface-normal n_y , volume-coordinate y , and wall-shear τ_y sign, and trained for 50 epochs with 8-GPU DDP for an effective batch size of 16. A notable empirical finding from SENPAI was the application of this lightweight STRING-inspired encoding from computer vision and robotics use cases to CFD surrogate modelling. Full run IDs and hyperparameters for the selected single model and ensemble are listed in Appendix E.

Caveat. Finally, a separate 50-epoch ensemble trained by SENPAI improved on all reported surface-pressure relative- L_2 metrics for DrivAerML (3.69%) and improved on AB-UPT for wall-shear relative- L_2 error (6.86%). A large gap in test-set volume-pressure relative- L_2 error was observed relative to AB-UPT and Transolver-3, potentially in part due to the shorter training time of SENPAI compared to the 500-epoch trainings of AB-UPT and Transolver-3. This training-duration discrepancy highlights an important decision criterion in autonomous research training: whether to scale up to longer, more expensive training runs after observing promising smaller-scale experimental results.

4.4. AirfRANS

Result. On the AirfRANS full benchmark, SENPAI, starting from a Transolver-based implementation, improves substantially on the original Transolver surface-MSE result (5-seed mean 0.0013 versus 0.0142) and on the strongest published surface reference, SpiderSolver (0.0043). The same model obtains mean volume-MSE of 0.00451, improving on GeoANF but remaining above SpiderSolver and slightly above the original Transolver volume result. SENPAI therefore achieves the strongest reported surface-MSE among these references, while the full benchmark remains limited by volume accuracy.

Table 2. AirfRANS full-task test MSE. Targets are train-statistic normalized, per-case means on the official test split; lower is better. Spider denotes SpiderSolver (Qi et al., 2025).

Metric	SENPAI	Spider	GeoANF	Transolver
Surface MSE	0.00130	0.0043	0.0089	0.0142
Volume MSE	0.00451	0.0017	0.0062	0.0037

Table 3. TandemFoilSet-Balanced 5-seed mean denormalized surface-pressure MAE. Lower is better.

Metric	MAE
Single-foil in-distribution	17.2905
Race-car camber holdout	31.2177
Cruise camber holdout	28.3145
Reynolds-number holdout	16.9834
Average	23.4515

Recipe and caveat. The final AirfRANS recipe SENPAI found used a 2-layer, 256-hidden-unit, 4-head SENPAI-Transolver with 96 slices, fixed multiscale Fourier coordinate features, a zero-initialized surface refinement head, Lookahead-wrapped AdamW, cosine scheduling, gradient clipping, weight decay, EMA with target decay 0.999, and a volume-weighted normalized-MSE objective with volume loss weight 10.0. All five seeds beat the published SpiderSolver surface-MSE result, while volume-MSE remains above SpiderSolver and more sensitive to seed. The complete five-seed configuration is reported in Appendix F.

4.5. TandemFoilSet

Result. Under the TandemFoilSet Experiment-4 task, SENPAI found a Transolver-based improvement on the cruise random uniform full-field MSE of the original paper ($1.7 \cdot 10^{-3}$ versus $1.0 \cdot 10^{-1}$). On TandemFoilSet-Balanced, SENPAI found a recipe that produced a denormalized surface-pressure MAE of 23.45 using a 5-seed average.

Recipe signal. The PR ledger shows the path: lower learning rates near $1e-4$, gradient clipping, EMA, warmup, and schedule-length tuning. A separate no-W&B TandemFoilSet-Balanced audit reached `test_avg/mae_surf_p` = 27.2125 in PR #1285, beating the 40.927 target by 13.7145, supporting the same schedule-and-throughput conclusion under a separate audit lane.

4.6. Experiment-Logging Ablation

Setup. To investigate the role of centralized experiment observability, an ablation compared SENPAI with hosted experiment logging against SENPAI using only locally written experiment results. Both settings used a 12-hour outer-

Table 4. Experiment-logging ablation on TandemFoilSet-Balanced. Surface-pressure MAE; lower is better.

Metric	Hosted log.	Local-only
Best test	44.15	27.21
Mean test	51.23	45.10
SD	4.79	12.89

Table 5. Behavioral differences in the experiment-logging ablation.

Behavior	With log.	No log.
Tool calls	60,600	78,372
Tool errors / 1k calls	63.10	32.23
Seed/variance refs. / 1k records	55.04	17.61
PRs opened	187	167
PR merge rate	31.0%	23.5%
Median result-to-merge latency	0.294 h	0.162 h

loop budget and a 30-minute per-experiment cutoff. SENPAI’s local-only logging trial achieved the strongest average final performance on TandemFoilSet-Balanced across five 12-hour trials, but with substantially higher variance, including one trial that failed to log a valid test-set result.

For the hosted-logging Willow arm, Figure 3 summarizes how hypotheses moved from Advisor branches through experiment families to final PR outcomes.

Behavior. Analyzing the Claude Code logs, centralized, queryable logging improved auditability and verification behavior. Agents with hosted logging more often verified run histories, compared split-level metrics, checked seed/control behavior, and reasoned about stale branches before accepting results. This produced a cleaner audit trail and more consistent provenance. The local-only arm was less verification-oriented, referencing keywords related to seeds and experimental variance roughly $3\times$ less. While the local-only arm achieved a strong average result, over longer experiment durations the performance of this setup is likely to drift.

4.7. Flat-Agent Comparison

Setup. To contrast SENPAI’s Advisor-mediated loop, we compared the performance of a *kagent* harness (Anonymus, 2026a) against the SENPAI harness using the TandemFoilSet-Balanced dataset and the same compute budget. *kagent* is a lighter-weight harness that drops the Advisor/Student split in favor of a flat Kaggle-like competition with a shared leaderboard; each agent keeps its own experiment journal and is prompted to compete aggressively to climb the leaderboard. In contrast to a Kaggle competition, the agents also have access to the same hosted experiment-logging project and commit history, so they can

Table 6. TandemFoilSet-Balanced 12-hour harness comparison. Lower average surface-pressure MAE is better.

Metric	SENPAI	<i>kagent</i>
Best avg-surf- <i>p</i> MAE	44.15	34.58

learn from other agents’ work. The *kagent* prompt and the full comparator run summary are provided in Appendix G.

Interpretation. In a direct 12-hour head-to-head, *kagent* behaved like a public Kaggle sprint: agents trained short variants, watched the leaderboard and shared hosted-logger configs, and quickly reused recipes that worked for peers. This fast peer diffusion, including copied hyperparameters and occasional prediction ensembles, likely gave it an exploitation-speed edge over SENPAI’s slower PR-mediated review loop. While this was successful in a 12-hour experiment setting, the inability to steer these agents and the less rigorous polling and monitoring infrastructure are likely to lead to drift and context rot as each Claude Code agent’s context window fills.

5. Discussion

5.1. Observable State as a Control Surface

Control surface. The PR and experiment-tracker ledger made SENPAI useful as a research assistant rather than a tool to execute experiments. Because hypotheses, code diffs, metrics, failures, and review decisions were all recorded in systems researchers already use, humans could inspect the research programme without needing to read the agents’ context. The same records also gave agents a stable system of record: they could query prior runs, identify stale experiments, compare seed behavior, and decide whether a new result actually advanced the baseline. In SENPAI, observability is therefore not simply logging; it is the layer through which both humans and agents can steer the research programme.

Recovery. This externalized state also reduced the fragility of long-running agent sessions. When agents compacted, restarted, or lost local scratchpad context, the next cycle could reconstruct the experiment from the PR description, code diff, git history, experiment-tracker records, and posted result comments. Local summaries remained useful for focus, but they were treated as caches over a more durable experiment record.

Research programme. The ledger also turned a collection of individual experiments into a queryable research programme. Researchers and agents could ask what had been tried, which failures were informative, which recipes transferred across datasets, where progress had stalled, and

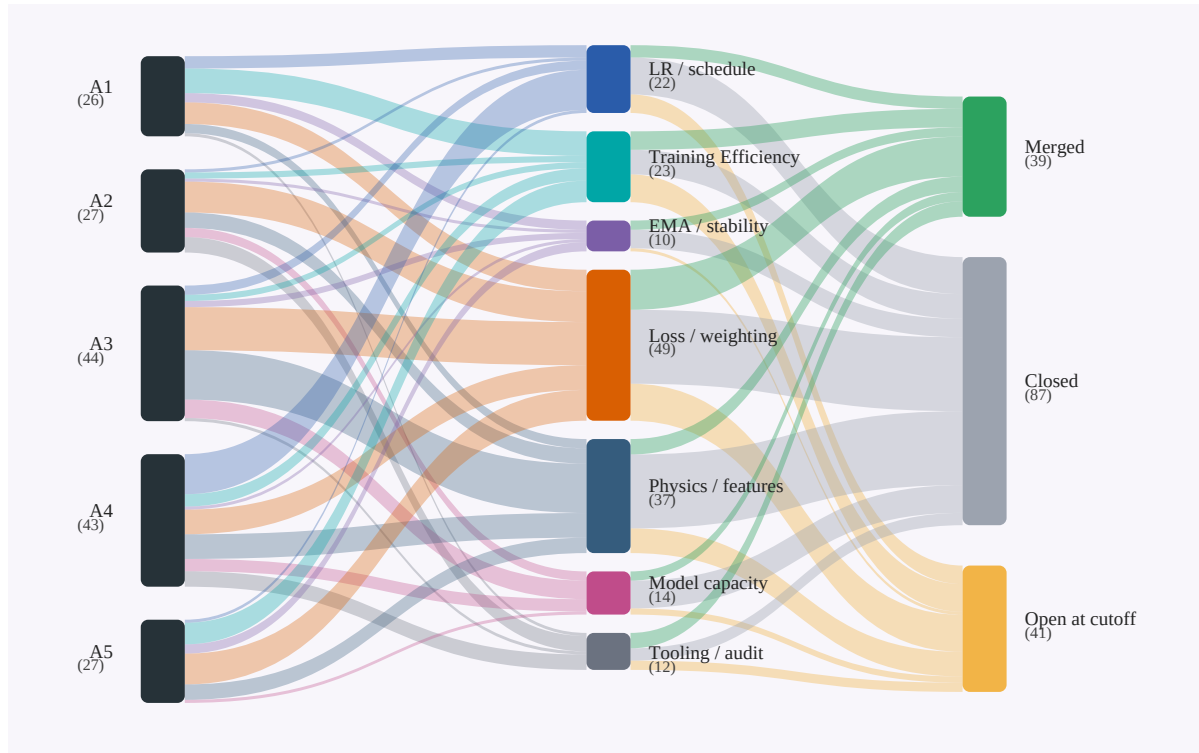


Figure 3. Experiment-ledger flow for the hosted-logging Willow ablation arm on TandemFoilSet-Balanced. Five Advisor branches (A1–A5) are routed through hypothesis families and into final PR outcomes; ribbon width is proportional to PR count. The figure compresses 167 GitHub-mediated experiments into a researcher-readable audit map, showing where autonomous search effort was allocated and how it resolved into merged, closed, or still-open work.

which resources were idle. This is SENPAI’s central systems claim: the goal is not to replace the researcher, but to increase the bandwidth at which researchers can understand, guide, and accelerate experimentation.

5.2. Harness Tradeoffs

Tradeoff. The kagent comparison highlights a harness-design tradeoff. Flat, leaderboard-driven cohorts can share successful recipes quickly because agents observe peer results directly and optimize for short-horizon score improvement. In contrast, SENPAI’s PR-mediated loop is slower but preserves review decisions, experiment provenance, and sparse human steering. We view kagent as useful for rapid exploitation, and SENPAI as better suited to longer-running programmes where auditability and recovery matter alongside score.

5.3. Failure Modes: Where Long-Running Research Agents Break

Observed failures. SENPAI’s most common failures were harness-engineering failures: monitoring long-running jobs, recovering across compaction or restarts, enforcing brittle tool interfaces, and reconciling PR state

with experiment-tracker state. They were less often failures of scientific reasoning. This distinction matters because it suggests that improving autonomous research systems requires better observable infrastructure, not only stronger prompts or more capable models. Additional artifact schemas and the extended taxonomy are listed in Appendix H.

Monitor bloat. During development, one dominant cost to running SENPAI was monitor-driven context bloat. In one notable experiment, Students monitored training logs for progress, errors, checkpoints, and completion, but persistent tail `-f | grep` events repeatedly resumed long-lived Student sessions for low-information log lines. In a 24-hour window with 57 Students running, 615 monitor setups produced 28,247 agent-facing monitor events and 28,246 agent responses, consuming 3.25B cache-inclusive tokens. The median agent response reloaded roughly 110k cached tokens while adding only 347 new uncached input tokens. This is an example of an architectural failure mode when running long-running research systems: training monitors should be cheap and escalation-based, surfacing only relevant events such as errors, experiment completion, or metric milestones.

Tool interfaces. Tool-use errors were smaller but systematic: in the same 24-hour window, 892 of 25,365 tool results had errors. The main causes were failed shell commands, GitHub scope errors, blocked wait patterns, wrong paths, failed pushes, oversized reads, and edit-guard failures. These errors point to a practical lesson: high-frequency harness operations should be executable helpers with narrow interfaces, rather than repeatedly reconstructed shell or GitHub operations inside agent context.

State drift. Compaction introduced a separate risk. The 24-hour window contained 240 automatic compactions across the 57 Students, with Student contexts reduced from roughly 196k tokens to 3k-token summaries. These summaries were sufficient for resumption, but lossy: state drift appeared in the agent conversation logs with mistakes such as incorrect GitHub labels or confusion over live experiment state. SENPAI’s experiment ledger mitigated this risk by externalizing much of the key experiment state. As a result, in that 24-hour window, despite an average of 4.2 compactions per Student, 38 of 39 Student sessions whose monitors showed training had completed also left PR comments as expected, and 33 of 39 PRs reached ready/status-review handoff as expected, indicating that despite multiple compactions, Students were able to use the experiment ledger’s external state to course-correct and take experiments to completion.

Mitigation. From observing these failure modes, we conclude that long-running research agents need robust, observable infrastructure more than additional prompt prose. Monitor bloat requires thoughtful observation; tool errors require executable helpers; training failures require mandatory result and checkpoint capture; state drift requires a single authoritative ledger, with summaries treated as caches. SENPAI did not eliminate failure, but it made failure measurable, attributable, and repairable.

5.4. Limitations

Scope. SENPAI is task-agnostic as a harness, but our empirical evidence comes from CFD-surrogate recipe search. We have not yet evaluated whether the same design transfers to domains with different data modalities, evaluation loops, or laboratory constraints.

Surface metrics. The strongest empirical results are on surface quantities. This matches many aerodynamic design workflows, where surface pressure and forces are primary signals, but volume-field prediction remains less developed; for example, our AirfRANS and DrivAerML experiments improve surface MSE while remaining behind the strongest published methods on volume MSE.

Supervision. SENPAI is semi-autonomous rather than fully autonomous. It requires an initial task repository, metric definition, dataset setup, and occasional human steering. The GitHub Issues and PR-comment interface is auditable and low bandwidth, but slower than interactive supervision.

Infrastructure dependence. Finally, SENPAI depends on mature external systems of record and substantial compute/API budget. In our implementation these were GitHub and Weights & Biases, which improved observability but also introduced dependence on hosted-service reliability, access control, rate limits, and retention policies. We will release the harness and experiment ledger so future work can audit costs, reproduce results, and evaluate more efficient variants.

6. Conclusion

SENPAI shows that a lightweight, observability-first harness can sustain semi-autonomous ML research when multi-agent state is grounded in the artifacts researchers already inspect: pull requests, git history, and experiment-tracker runs. Across a multi-week CFD-surrogate research programme, this design supported thousands of autonomous experiments over three benchmarks while allowing researchers to steer the system through ordinary PR review and a GitHub Issues channel.

The main lesson is that reliability comes from the harness, not the agent instructions alone. Long-running research agents did not primarily fail because they lacked clear instructions; they failed at system boundaries: brittle tool interfaces, unbounded monitoring, missing checkpoint capture, and drift between derived summaries and the authoritative ledger. Prompt hardening alone did not remove these failure modes. Durable external state, clear scripts for passing work between agents, filtered monitoring, and mandatory result capture were more important for keeping the research programme recoverable and auditable.

SENPAI’s contribution is therefore not to replace the researcher, but to increase the bandwidth at which researchers can understand, guide, and accelerate autonomous experimentation. By making both the inner experiment loop and the outer harness loop observable through mature systems of record, SENPAI turns autonomous research from an opaque agent workflow into a queryable, reviewable research programme. We will release the harness, experiment ledger, and failure-mode analysis to support future work on more reliable and efficient autonomous research systems.

Software and Data

For double-blind review, non-anonymous links to the harness repository and hosted run ledger are withheld from

this version and will be included in the camera-ready version. The CFD datasets used in the experiments are DrivAerML, AirFRANS, and TandemFoilSet; access and redistribution follow the licenses and terms provided by the original dataset authors.

Impact Statement

This work aims to make autonomous ML research more auditable, recoverable, and steerable by grounding agent activity in systems of record that researchers already inspect. The positive impact is increased researcher bandwidth for expensive scientific ML experimentation, especially in domains where strong baselines require repeated empirical iteration. The main risks are increased compute/API consumption, over-trusting autonomous experiment choices, and propagating errors when agent-written summaries drift from measured evidence. SENPAI mitigates these risks by keeping humans in the review loop, preserving experiment provenance in PRs and tracked runs, and treating summaries as caches rather than authoritative state.

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A. Agent Instructions and Source Map

The experiments in this paper were controlled by a small number of versioned text artifacts rather than by hidden agent memory. The exact prompt files will be released with the SENPAI harness; this appendix summarizes their operational content so that the harness can be audited without reprinting every instruction verbatim.

Table 7. Primary instruction artifacts used by the harness and comparator.

Artifact	Location	Role in the system
Advisor prompt	<code>system_instructions/CLAUDE-ADVISOR.md</code>	Defines the no-GPU research lead: survey PRs, review result comments and W&B runs, merge validated winners, close dead ends, and assign one-hypothesis PRs to idle Students.
Student prompt	<code>system_instructions/CLAUDE-STUDENT.md</code>	Defines the GPU worker: claim only assigned PRs, edit the task training file, run bounded experiments, report exact commands, metrics, W&B IDs, and structured terminal results.
Researcher sub-agent	<code>.claude/agents/researcher-agent.md</code>	Defines a literature and experiment-history specialist used to produce mechanism-grounded experiment proposals and research-state updates.
Problem contract	<code>DrivAerML/program.md</code>	Example task contract specifying mission, data, editable files, metric keys, checkpoint selection, telemetry, and paper-facing target values.
TandemFoil2 contract	<code>target/tandemfoil2/program.md</code>	Active local problem contract for the kagent-seeded TandemFoil2 package, including split names, edit boundaries, and the <code>val/12_error</code> leaderboard metric.
Flat comparator prompt	<code>KAGGLER_AGENT.md</code>	Defines the kagent loop: watch leaderboard and W&B, edit local training/prediction code, keep only useful code changes, always journal outcomes, and continue until interrupted.

The most important verbatim prompt element is the terminal result marker, which makes a Student’s PR comment machine-checkable:

```
SENPAI-RESULT: {"terminal":true,"status":"complete","pending_arms":false,
"wandb_run_ids":["<run-id>"],"primary_metric":{"name":"<metric>","value":<number>},
"test_metric":{"name":"<metric>","value":<number>}}
```

The Advisor prompt additionally contains a plateau protocol: after five or more non-improving experiments, the Advisor must escalate from local tuning toward a different strategy tier, revisit first principles, use the researcher sub-agent, and propose larger mechanism changes rather than treating the plateau as a stopping signal.

B. Problem-Program Contract

Each SENPAI target repository carries a `program.md` file that makes task assumptions explicit. The DrivAerML contract used as a reference artifact defines the elements in Table 9; the local TandemFoil2 contract follows the same pattern, with `val/12_error` averaged over four validation splits as its leaderboard metric. These files are intentionally more specific than generic prompts: they are the task-local source of truth for what a valid experiment may edit, how data are structured, and which metrics count as paper-facing evidence.

C. Skill Catalogue

SENPAI’s prompts route common operations through small helper skills and a shared `senpai-gh` library. This was a deliberate control choice: high-frequency GitHub and bookkeeping operations should have narrow, testable interfaces instead of being reconstructed from scratch by each agent invocation.

D. Advisor Outer Loop

Table 8. Role contracts induced by the prompts.

Role	Inputs	Allowed actions	Required output
Advisor	PR ledger, W&B runs, Student pod state, human GitHub issues, and <code>program.md</code> .	Create draft PRs, assign labels, review each completed PR, request revisions, close dead ends, and merge winners after structured terminal results.	Updated baseline and research-state files, PR review comments, and new hypothesis PRs for idle Students.
Student	Assigned PR, target branch, <code>program.md</code> , training logs, W&B runs, and Advisor comments.	Modify the task training file, run the assigned experiment, debug crashes/OOMs, and resubmit after requested changes.	A PR comment beginning with a single-line SENPAI-RESULT JSON marker, followed by metrics, run IDs, commands, interpretation, and follow-ups.
Researcher	Prior PRs, W&B evidence, code lineage, papers, repos, and domain references.	Search literature and experiment history, reconstruct mechanisms, and design discriminating follow-up experiments.	A decision-useful synthesis: mechanism, evidence, implementation notes, proposed experiment tree, stop condition, and confidence.
kagent	Shared leaderboard, own experiment journal, W&B configs, visible commits, and local logs.	Iterate directly on <code>train.py</code> and <code>predict.py</code> , run training/prediction, retain or discard code, and write the journal after every trial.	Scored submissions plus a persistent experiment journal, with no Advisor review gate.

The key property is that ordinary polling remains outside the LLM. The Advisor is invoked only when there is actionable state: a PR to review, a human issue to answer, or idle Students that need assignments. The watermark advances only after successful invocations, so transient failures retry against the same triage state.

E. DrivAerML Recipe Details

The `string_multisigma` encoding is a lightweight adaptation of separable translationally invariant position encodings (Schenck et al., 2025): per-axis learnable sine/cosine bases are initialized across several spatial scales and projected into the Transolver width.

The ensemble result in Table 1 is documented in PR #602. It was a greedy forward-selection ensemble over test predictions from seven independently trained checkpoints: `nh96x7m4`, `5o7jc7wi`, `wyz68o8r`, `9mm3sz7x`, `49aimdiz`, `19qf6di1`, and `nh2ke150`. Candidates were added one at a time to minimize validation field error, and the final inference-time prediction averaged the selected members uniformly.

F. AirfRANS Recipe Details

G. kagent Comparator Details

The kagent comparator used a flat competition prompt rather than SENPAI’s Advisor/Student split. Its prompt instructs each agent to read the task README and experiment journal before coding, check the leaderboard and W&B before each iteration, formulate a hypothesis, modify only the training/prediction entypoints, train and predict, keep code only when it improves, and always append an experiment-journal entry even for failed or discarded trials. The primary metric was validation L2 error, with memory pressure, turbulence, and no-slip boundary conditions called out as domain challenges. Unlike SENPAI, this loop is explicitly autonomous once started and does not pause for Advisor review.

TandemFoilSet parity run. An eight-agent kagent cohort ran on the TandemFoilSet parity benchmark from 2026-04-23 15:53 UTC to 2026-04-24 04:00 UTC, using one GPU pod per agent, a 30-minute per-training-run budget, and the Transolver starter. The run produced 330 agent commits and 223 scored leaderboard updates without human intervention. Agents communicated only through shared data/prediction volumes, the git remote and leaderboard, and W&B telemetry.

The most important qualitative finding was rapid peer diffusion. Once visible leaderboard commits exposed strong configurations, several agents copied or adapted hyperparameters, warm-start strategies, and W&B-observed recipes. The decisive improvement was a full-mesh fine-tuning unlock: several top agents concluded that training on heavily subsampled point

Table 9. Information supplied by a task `program.md`.

Contract component	Example content in the DrivAerML programme
Mission and targets	Minimize held-out <code>test_primary/*</code> metrics for surface pressure, vector wall shear, and volume pressure, using validation for steering but final test metrics for claims.
File boundaries	<code>train.py</code> , <code>model.py</code> , and runtime helpers are experiment-editable; data loaders, manifests, and split files are read-only during normal experiment PRs.
Data contract	Surface tensors include coordinates, normals, area, pressure coefficient, and wall-shear components; volume tensors include coordinates, signed-distance field, and volume pressure.
Split and evaluation	The public processed split is fixed at 400 train, 34 validation, and 50 test cases, with deterministic full-fidelity evaluation chunks for validation and test.
Model interface	The reference model consumes padded surface and volume tensors plus masks, and must not compute losses or metrics on padding tokens.
Telemetry and validity	W&B must contain finite final primary metrics, plus gradient, weight, slope, validation, and test diagnostics sufficient to diagnose unstable or invalid runs.
Paper-facing metrics	Individual target-family relative- L_2 percentages are reported; the aggregate checkpoint metric is treated as a triage scalar rather than a benchmark column.
Workflow	Roles are coordinated through PRs and GitHub issues; final result claims must be traceable to run IDs and code state.

Table 10. Helper skills used by the Advisor and Students.

Helper	Caller	Effect on the experiment ledger
<code>survey-prs</code>	Advisor	Reads review-ready, WIP, draft/stalled, and idle-Student state for an Advisor branch.
<code>poll-for-work</code>	Student	Returns assigned WIP PRs for a Student label and reads comments to detect revisions.
<code>check-human-issues</code>	Both	Finds labeled human issues for the caller and posts role-prefixed replies without closing issues.
<code>list-experiments</code>	Researcher	Exports merged winners, compact result tables, and detailed PR bodies from the PR ledger.
<code>assign-experiment</code>	Advisor	Creates a Student branch, opens a draft PR, labels it <code>status:wip</code> , and writes full experiment instructions.
<code>submit-experiment-results</code>	Student	Commits changed experiment files, pushes the branch, marks the PR ready, and swaps <code>status:wip</code> to <code>status:review</code> .
<code>merge-winner</code>	Advisor	Preflights terminal results, squash-merges a winning PR, updates the Advisor baseline, and pulls the updated branch.
<code>senpai-gh</code>	Both	Provides lower-level label swaps, PR comments, closure, preflight, and read-only query primitives.

sets created a train/eval mismatch for attention layers that pool over nodes, and moved from subsampled pretraining to batch-size-2 full-mesh fine-tuning. Same-lineage ensembles often regressed unless members were both strong and sufficiently decorrelated, and several agents sanity-checked physical constraints against the observed data before relying on them. A separate scorer-race incident marked partially written prediction directories as terminally incomplete, illustrating the same infrastructure-contract fragility discussed in Appendix H.

External competition run. As an additional stress test, sixteen Claude Opus 4.6 kagent agents were pointed at the GRaM ICLR 2026 open competition from 2026-03-29 to 2026-03-30. The winning cohort member reached 0.8526 mean L_2 against a roughly 1.75 reference-MLP baseline, and the strongest checkpoint was packaged and submitted to the upstream competition as a live pull request. This run predates the later experiment-journal convention, so its evidence is primarily leaderboard and commit based.

H. Evidence Artifacts and Failure Taxonomy

The experiment ledger supports several supplementary audit views beyond the main tables:

- full benchmark result tables for AirfRANS, TandemFoilSet parity, TandemFoilSet-Balanced, and DrivAerML validation/test trajectories;

Algorithm 1 Advisor triage loop outside the LLM context

```

1:  $iteration \leftarrow 0$ ;  $last\_check \leftarrow \emptyset$ 
2: while true do
3:    $iteration \leftarrow iteration + 1$ 
4:    $review\_ready \leftarrow \text{LISTREVIEWREADYPRs}(branch, last\_check)$ 
5:    $human\_issues \leftarrow \text{LISTNEWHUMANISSUES}(branch, last\_check)$ 
6:    $idle\_students \leftarrow \text{LISTIDLESTUDENTS}(students, branch)$ 
7:   if  $iteration > 1$  and  $review\_ready$ ,  $human\_issues$ , and  $idle\_students$  are empty then
8:     sleep for one harness tick and continue
9:   end if
10:   $message \leftarrow \text{RENDERTRIAGE}(review\_ready, human\_issues, idle\_students)$ 
11:  if  $iteration = 1$  then
12:     $exit\_code \leftarrow \text{RUNAGENT}(role\_prompt + message)$ 
13:  else
14:     $exit\_code \leftarrow \text{RESUMEAGENT}(heartbeat + message)$ 
15:  end if
16:  if  $exit\_code = 0$  then
17:     $last\_check \leftarrow \text{MAXUPDATEDAT}(review\_ready, human\_issues)$ 
18:  end if
19:  sleep for one harness tick
20: end while

```

Table 11. Selected DrivAerML SENPAI single-model configuration.

Setting	Value
Source	PR #741 in the experiment ledger
W&B runs	train 1szc4ri7, eval 1tal40wr
Training budget	50 epochs; validation-selected checkpoint
Parallelism and batch	8-GPU DDP; batch size 2 per rank, 16 global
Optimizer	Lion, learning rate 1e-4, weight decay 1e-4
EMA	0.999
Backbone	4 layers, 512 hidden units, 4 attention heads, 128 slices
Positional encoding	string_multisigma; sigmas 0.25, 0.5, 1.0, 2.0, 4.0
Point sampling	Train: 65,536 surface / 65,000 volume; eval: 40,000 surface / 65,000 volume
Augmentation	Training-only y -plane reflection with lateral coordinate, normal, and τ_y sign flips
Loss	Normalized MSE with surface/volume weights 1.0/1.0

- an experiment-ledger CSV schema with PR number, title, status, Student, W&B run IDs, primary metric name/value, baseline delta, era tag, and merge-decision timestamp;
- redacted examples of real PR conversations linked to W&B panels;
- sprint evidence logs including experiment snapshots, run panels, advisor-branch git logs, and urgent human issue timelines.

We organize full failure-mode notes under four infrastructure-facing categories. *Monitoring failures* include unfiltered log monitors, heartbeat events that wake agents for low-information lines, and long-lived streams that inflate cached context. *Tool-interface failures* include brittle shell/GitHub sequences, missing scopes, wrong paths, blocked waits, failed pushes, oversized reads, and edit-guard failures. *Result-capture failures* include missing final metrics, invalid terminal result markers, checkpoint selection mistakes, and scorer/checkpoint races. *State-reconciliation failures* include stale PR labels, incomplete W&B-to-PR reconciliation, lossy compaction summaries, and conflicts between markdown caches and the authoritative ledger.

These categories explain why SENPAI’s mitigations are mostly infrastructural: filtered monitors, narrow helper scripts, terminal result schemas, checkpoint preflights, and treating local summaries as caches over the PR/git/W&B ledger.

Table 12. AirfRANS SENPAI configuration from PR #3135.

Setting	Value
Source	PR #3135, AirfRANS: EMA=0.999 + volume-loss-weight=10x compound
Base and head branch	radford; nezuko/af-ema-plus-vol-weight
Merge commit	cc25dc2e1de415bb178d1943df8573f382bd07d4
Original W&B run	sh2zyfwr
Five-seed W&B group	af-champion-5seed-20260506
Dataset/task	AirfRANS full
Optimizer and schedule	Lookahead-wrapped AdamW, learning rate 6e-4, cosine schedule with T_max=50
Regularization	Weight decay 1e-2, gradient clipping 1.0, EMA target decay 0.999
Backbone	2 layers, 256 hidden units, 4 heads, 96 slices, MLP ratio 4
Positional features	Fixed Fourier coordinate features at frequencies 0.5, 2.0, 8.0, and 32.0
Surface refinement	Enabled; zero-initialized 2-layer MLP with 128 hidden units
Sampling	8,000 surface anchors, 8,000 volume anchors, 25,000 geometry points, 4,096 supernodes
Loss	Normalized surface MSE plus 10.0× normalized volume MSE

Table 13. AirfRANS five-seed verification runs.

Seed	W&B run	Selected epoch	Surface MSE	Volume MSE
42	e1761ghv	525	0.000694	0.004048
123	m02h16z6	268	0.001998	0.008007
789	fz1lvy3i	687	0.001786	0.003769
2026	y41sww82	737	0.000995	0.003142
3135	p5xe5cd3	577	0.001029	0.003585
Mean	—	—	0.001300	0.004510
Median	—	—	0.001029	0.003769

Table 14. kagent April 23 TandemFoilSet parity leaderboard. Surface-pressure MAE; lower is better.

Rank	Agent	Avg.	Single in-dist.	Geom. RC	Geom. cruise	Re holdout
1	frieren	34.41	38.73	47.92	18.21	32.78
2	fern	42.62	46.80	55.53	26.65	41.50
3	edward	44.00	48.00	59.19	26.44	42.35
4	askeladd	48.89	52.09	62.46	32.02	49.01
5	alphonse	56.21	57.41	70.36	40.80	56.29
6	thorfinn	62.57	45.81	77.35	40.44	86.70
7	tanjiro	68.25	55.52	84.19	45.00	88.27
8	nezuko	85.84	96.56	96.43	63.43	86.94

Table 15. Median token load observed per PR in the rough appendix evidence log.

Metric	Value
Median Student input tokens per PR	1,421,948
Median Student output tokens per PR	6,012
Median Advisor input tokens per PR	187,453
Median Advisor output tokens per PR	1,621